Vehicle Task Assignment using Group VIKOR

Introduction

This project presents an intelligent vehicle-task assignment system using the Group VIKOR multi-criteria decision-making method in Python. The system evaluates multiple vehicles based on different criteria and automatically assigns the most suitable vehicle for specific tasks.

The project combines:

- Multi-Criteria Decision Making (MCDM)

- Group VIKOR ranking

- Task-specific weighting

- Data analysis and visualization

- Automated Excel report generation

Dataset

The project uses a vehicle dataset adapted from Kaggle datasets.

The dataset contains vehicle information such as:

- Price

- Mileage

- Engine Capacity

- Manufacturing Year

- Transmission

- Fuel Type

Tasks

The system evaluates vehicles for different tasks:

- City Delivery

- Long Distance Travel

- Budget Purchase

- Performance Driving

- Modern Family Car

Each task uses different importance weights for the evaluation criteria.

Criteria

The following criteria are used in the decision-making process:

Price = Vehicle purchase cost

Mileage = Vehicle usage distance

Engine Capacity =Vehicle performance

Year = Vehicle age and modernity

Methodology

The Group VIKOR method is used to rank vehicles according to multiple criteria.

The process includes:

1. Data preprocessing

2. Criteria normalization

3. Task-specific weight assignment

4. Calculation of S, R and Q values

5. Vehicle ranking

6. Best vehicle assignment per task

-Lower Q scores indicate better alternatives.

Results

The system generates:

Best vehicle assignments, Full vehicle rankings, Task-specific weight analysis, Q-score visualizations, Excel reports with charts

The generated charts help visualize:

Criteria importance, Vehicle ranking performance, Task-based decision preferences

Technologies Used

Python, Pandas, NumPy, Matplotlib, XlsxWriter

Conclusion

The project successfully demonstrates how Group VIKOR can be applied to intelligent vehicle-task allocation problems.

Different tasks require different vehicle characteristics, and the system dynamically adapts decision weights to select the most appropriate vehicle for each scenario.

This approach can be extended to:

Fleet management, Logistics optimization, Smart transportation systems, Resource

allocation problems

References

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